

**Voting Power Beyond Formal Authority: A Comparative
Analysis of the United Nations Security Council,
International Monetary Fund, and India's Goods and
Services Tax Council**

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Abstract

The objective of this research is to examine the power allocation in three significant international organisations, the UN Security Council, the International Monetary Fund, and the Goods and Services Tax Council of India, utilising cooperative game theory in order to explain the concept of voting power in practice. In the research, the author employs the Banzhaf and Shapley-Shubik Power Indices to convert the rule of voting into the framework of a weighted voting game in order to reveal the gap between the number of votes and power.

The theoretical foundation of this study has been developed in a series of academic contributions, ranging from the empirical findings of Banzhaf concerning the disproportionate relation between votes and power dating back to 1965, up to the characterisation of weighted voting by Taylor and Zwicker.

In all three cases, the study highlights the significant disjunctures between authority and power. In the UNSC, the veto structure makes it extremely powerful, as each permanent member enjoys 16.69% of Banzhaf power, despite equal representation with other members who have only 1.65% of the same. The IMF shows that swing voters are highly powerful, as the "Rest of the World" group enjoys 50% of power, even while having only 41.85% of the vote share. The GST Council is the most imbalanced case, where the Centre has 89.74% of Banzhaf power (37.5% under the Shapley-Shubik method), despite being allocated just one-third of the vote share constitutionally.

It has been demonstrated that voting systems are mathematical objects whose properties, including weights, quota, threshold, and veto structure, influence the power distribution, irrespective of the intentions of their designers. This research has implications regarding legitimacy, accountability, and institutional reforms. The understanding of how multinational institutions function should include rigorous mathematical analysis.

Introduction

The mathematical study of voting power occupies a peculiar position at the intersection of pure theory and applied institutional analysis. On its surface, the problem appears straightforward: given a set of voters and a decision rule, how much influence does each voter actually wield? Yet beneath this deceptive simplicity lies a profound disconnect that has animated decades of scholarship the distinction between formal voting weight and effective political influence. A legislator holding fifty votes in a hundred-member body possesses vastly greater formal authority than one holding a single vote, yet may exercise zero pivotal power if a competing coalition of fifty-one votes can pass any resolution regardless of their support or opposition. Conversely, a voter holding seemingly modest weight may prove indispensable to nearly every winning outcome, commanding political influence wildly disproportionate to their formal allocation. This paradox, first systematically exposed by John Banzhaf in 1965, reveals that voting systems cannot be understood through weight assignments alone. They must be analysed as mathematical objects whose internal structural properties determine the actual distribution of decision-making authority.

Research Objective and Questions

This dissertation investigates a deceptively simple yet institutionally consequential question: What is the actual distribution of decision-making power in multilateral institutions, and how does it diverge from their formal constitutional rules? Rather than assuming that voting power flows directly from vote allocation, this research applies formal cooperative game theory specifically the Banzhaf Power Index and Shapley-Shubik Index to three significant international and domestic institutions to reveal the structural properties that determine real influence.

The research is guided by four interrelated questions: how the structural features of weighted voting systems, such as vote weights, decision thresholds and veto mechanisms shape the distribution of effective decision-making power and generate divergences between nominal authority and actual pivotal power; how the decision-making structures of the United Nations Security Council, International Monetary Fund, and GST Council can be modelled as weighted voting games under their respective constitutional rules, and whether common patterns emerge across these institutions

operating in distinct governance contexts; how the distribution of voting power among member states or constituent actors, as measured by the Banzhaf Power Index and the Shapley–Shubik Index, compares within and across these bodies; and what the implications of such divergences are for institutional legitimacy, democratic accountability, and the design of multilateral decision-making systems.

Case Selection and Comparative Framework

To answer these questions, this research examines three consequential multilateral institutions: the United Nations Security Council (UNSC), the International Monetary Fund (IMF), and India's Goods and Services Tax Council. The choice reflects a deliberate methodological strategy. First, each represents a distinct species of weighted voting architecture: the UNSC employs an explicit veto mechanism concentrated among five permanent members, the IMF distributes voting shares proportionally according to economic capacity and the GST Council combines a supermajority threshold with two-tier federal representation. Second, each institution operates within genuinely consequential domains international peace and security, global monetary governance and India's fiscal federalism making formal analysis of voting power not merely a mathematical exercise but substantive investigation into how collective decisions are actually made. Third, their constitutional provisions are sufficiently well documented and stable that formal modelling is tractable, permitting precise computation rather than speculation. Together these cases enable comparative analysis illuminating both universal patterns in how weighted voting systems distribute power and institution-specific asymmetries arising from different constitutional designs.

Theoretical Foundation

The theoretical foundation rests on three seminal contributions establishing that voting power is fundamentally distinct from voting weight. Banzhaf's operational definition power as the ability to change the outcome of a vote shifted analytical focus from nominal vote allocation to the marginal contribution of each voter across all possible coalitions. His empirical demonstrations proved devastating to the intuitive assumption that proportional vote allocation yields proportional political influence. Taylor and Pacelli developed the theoretical framework characterising which voting

systems are representable through weights and quotas. Taylor and Zwicker completed this arc by proving that trade robustness constitutes the precise mathematical criterion distinguishing weighted systems from all others. These three strands collectively established that voting systems must be studied as formal objects subject to rigorous mathematical analysis if their actual functioning is to be understood.

Research Contribution and Significance

This research contributes across three dimensions. To formal voting theory, it demonstrates how power indices illuminate structural properties of institutional design. To comparative institutional analysis, it provides a common analytical framework for examining three otherwise disparate bodies. To specific policy debates surrounding UNSC reform, IMF governance, and India's fiscal federalism, it provides precise empirical measures of voting power distribution against which proposed reforms can be evaluated. The findings carry implications extending beyond formal analysis: they demonstrate that institutional legitimacy and formal rule design cannot be divorced from actual power distribution. When nominal voting weights and pivotal power diverge sharply, as they do in all three institutions examined, questions arise about whether voting systems in practice align with institutions' stated purposes and foundational principles.

This research proceeds from the conviction that mathematical rigour and institutional realism are complementary rather than antagonistic. The elegant formal properties of voting systems, properly understood, reveal truths about how power actually operates within institutions, thereby informing both scholarly understanding and reform advocacy.

Literature Review

Weighted voting systems represent a complex intersection of mathematics, political science, and institutional design. These systems assign numerical weights to individual voters or voting units, allowing decision-making bodies ranging from supranational organisations to municipal governments to differentiate voting power based on various criteria such as population, economic capacity, or political status. Despite their apparent mathematical elegance and theoretical appeal, scholarly investigation has revealed profound paradoxes underlying their implementation. This literature review examines three seminal contributions that collectively demonstrate a critical disconnect between the formal allocation of voting weights and the actual distribution of political influence. By synthesising the empirical critique offered by Banzhaf, the theoretical framework developed by Taylor and Pacelli, and the mathematical characterisation provided by Taylor and Zwicker, this analysis illuminates both the limitations of weighted voting systems and the structural properties necessary for their coherent operation.

Historical Context: The Constitutional Imperative

It should be noted that the appearance of the weighted voting technique as an instrument of policy was conditioned by a certain historical context in the legal history of America. In particular, after a series of famous Supreme Court rulings—*Baker v. Carr* (1962), *Gray v. Sanders* (1963), *Wesberry v. Sanders* (1964), and *Reynolds v. Sims* (1964), American state legislatures found themselves under the constitutional obligation of equalising constituency populations as much as possible. The Supreme Court's demand for proportional representation created some serious logistical difficulties in terms of policy: the impossibility of preventing gerrymandering, the necessity of preserving current administrative districts, regular adjustments, and a large legislative body resulting from all districts' individual representation.

In this sense, weighted voting looked like a more pragmatic solution compared to the redrawing of all constituencies. It allowed keeping the current (unequal) constituencies and giving representatives a number of votes proportional to their constituency populations. This technique seemed to balance

the constitutional requirements of equality with pragmatic needs of administration. Nevertheless, it was based on a certain assumption which remained unnoticed but crucial in essence: proportionality of voting led to proportionality of political decision making.

Banzhaf's Empirical Critique: Measuring Voting Power

However, according to the work of John F. Banzhaf III, written in 1965, the idea of voting power being associated with the number of votes was seriously questioned. Specifically, the core argument made by Banzhaf suggested that voting power was completely unrelated to the number of votes controlled by a particular individual. In addition, Banzhaf offered a significant definition: voting power is the ability to change the results of a vote and affect the adoption or failure of a measure by means of casting a vote.

The operational definition of power, in this context, was presented by Banzhaf as counting the voting combinations in a voting body when one's vote makes a difference in the results. The difference between such an approach and the vote counting method is that a representative with 49 votes out of 100 has zero power when another representative controls 51 votes because his/her votes have no effect on any outcome. However, in a situation when any combination of three representatives adopts the law but smaller combinations do not have this capability, then, irrespective of whether one controls 3, 50, or 100 votes, he/she has a significant power.

In Banzhaf's first case, there was a hypothetical legislature consisting of five legislators whose populations were 50,000, 10,000, 10,000, 10,000, and 10,000 people. Each of these legislators was entitled to five, one, one, one, and one vote respectively. As such, in order to pass legislation, at least five votes were required. This means that only the representative having five votes had the ability to form winning coalitions, while the other four were powerless despite representing 40% of the population.

The second hypothetical legislature consisted of four districts of 80,000 people each and another district of 10,000, which gave them eight votes each. Since legislation requires 17 votes out of 33,

Banzhaf's systematic calculation of all the possible 32 voting combinations showed that there was no difference between a weighted system and an unweighted one. The complicated mathematical equation was "an empty gesture" according to Banzhaf.

Apart from theoretical examples, Banzhaf used his technique on concrete examples of government systems. The weighted system in Nassau County, which gave nine votes to each of the towns Hempstead I and Hempstead II, seven votes to North Hempstead, three votes to Oyster Bay, and only one vote each to Glen Cove and Long Beach, showed a complete imbalance in the distribution of power among different municipalities. Only three municipalities had power to vote, while others were only symbolic voters in the sense that they did not have representation.

Banzhaf acknowledged important methodological constraints. His power index presupposed voter independence and equal probability across all voting patterns, an assumption he explicitly noted made no empirical claims about actual legislative behaviour, coalition formation, or political calculation. While his measure provided a necessary condition for constitutional adequacy, it could not capture all dimensions of representative authority. Nevertheless, the core insight remained revolutionary: proportional vote allocation constitutes neither necessary nor sufficient condition for proportional distribution of actual political influence.

Taylor and Pacelli: Theoretical Framework and Structural Properties

The theoretical groundwork required to comprehend when a voting system is or is not representable by weights and quotas was established by Taylor and Pacelli. The theoretical construction started with some rigorous definitions. Yes-no voting systems are sets of rules that define which combinations of votes result in the passing of a measure, essentially determining the choice between an option and the status quo. Taylor and Pacelli defined a coalition, which is any combination of individuals who make up the winning coalition if their unanimous vote ensures the measure's passage; all other coalitions are considered losing coalitions.

Weighted systems are defined in terms of existence of real-valued weight allocations and quotas, such that winning coalitions are exactly those whose combined weights are greater than or equal to the quota. In this manner, majorities, super-majorities, and the proportional representation systems criticised by Banzhaf are all included within this category. The theoretical approach adopted by Taylor and Pacelli relied heavily on the use of four specific examples of yes-no voting schemes: The weighted scheme of the European Economic Community, the veto power of the United Nations Security Council, the US federal legislative process, and the Canadian constitutional amendment process.

The authors of swap robustness have proposed that a system is considered to be swap robust if upon replacement of a particular voter from one winning coalition with a voter from another winning coalition whose membership is different, at least one coalition will still remain winning. All weighted systems are inherently swap robust but the reverse is not true. For instance, the U.S. federal system, which requires either 218 members of the House along with 51 Senators who agree on actions with the President, or 218 members of the House and 50 Senators acting with the President and the Vice President, or 290 members of the House with 67 Senators irrespective of the president, does not have swap robustness.

In a trade robust system, there exists an ability to exchange among several winning coalitions consisting of different groups of voters in one step. The system is said to be trade robust if trading cannot make all the previously winning coalitions become losing coalitions. All the weighted systems are trade robust but not all the trade robust systems are weighted.

Taylor and Zwicker: The Characterisation Theorem

Taylor and Zwicker's 1992 work represents the culmination of decades of investigation into weighted voting structures. Their fundamental contribution establishes mathematical equivalence between an algebraic property (representability through weights and quota) and a combinatorial property (trade robustness). Rather than employing convex geometry and polytope separation

approaches utilised in earlier characterisations Taylor and Zwicker developed reasoning intrinsic to voting theory itself.

The authors formalised k-trade robustness: a voting game exhibits this property if converting any sequence of k winning coalitions into losing coalitions through bilateral voter trades proves impossible. Significantly, coalitions in sequences need not be distinct, a refinement that strengthens the concept. Their central theorem establishes that for any voting game with n voters, three conditions are equivalent: (1) the system constitutes a weighted structure, (2) the system is fully trade robust, and (3) the system is $2^{(2^n)}$ -trade robust. The power of condition (3) lies in its computational feasibility. While examining all possible trade sequences remains infinite, verifying $2^{(2^n)}$ trade robustness though exponentially explosive is finite. For legislatures with ten or fewer members, computer enumeration becomes practical.

The characterisation theorem produces several immediate implications. First, it furnishes a finite procedure for determining whether real voting systems are weighted. Second, it clarifies non-weighted systems' structure: the United States federal system and Canadian constitutional procedure both exhibit specific trade sequences demonstrating non weightiness. Third, trade robustness emerges as the precise mathematical property capturing a system's representability through weights and quota.

Taylor and Zwicker's methodological approach merits emphasis. By grounding characterisation in combinatorial voting properties rather than geometric externalities, their work demonstrates that voting systems possess deep internal structural constraints. The equivalence between algebraic weightiness and combinatorial robustness reveals profound truths about how voting rules determine actual decision-making possibilities.

Synthesis and Interconnections

It may be said that the three papers are interdependent and form a theoretical framework. First, Banzhaf demonstrated in practice that the weight system does not provide the distribution of power according to the allocation of votes. Taylor and Pacelli developed the concept theoretically and

Taylor and Zwicker provided a mathematical basis for it. Thus, in the case described by Banzhaf, when the power belongs to only one member of parliament, the criterion of trade robustness is violated since the trades made may generate new winning coalitions that were impossible before. However, it should be noted that there is an important difference between the approach taken by Banzhaf and Taylor & Zwicker. The former estimates the power of representatives in certain systems, and evaluates the power of each player in relation to others. The latter consider whether a given system can be characterised as a weight system or not. In other words, the former approach is intra systemic, and the latter is a system level question.

Limitations, Assumptions, and Unresolved Questions

Although the models are theoretically sophisticated, they make some key assumptions that provide areas for further research. The assumption of voter independence where any voting behavior has equal probability is at odds with empirical legislative practices. Party identification, ideologies, and geographical interests produce highly correlated voting behaviors, making the assumption made by Banzhaf unrealistic. A more sophisticated model would consider the actual patterns of voting.

The lack of justification for why trades represent normative objections to non-weightedness must be considered. Although Taylor and Zwicker provide examples based on sports that demonstrate that systems in which trades are not possible should allow trades, such a claim is not formally justified. Why should there be a requirement of trade robustness for weighted voting systems?

Moreover, the notion that voting is the sole domain of political power is an over-simplification of institutional politics. The setting of the agenda, rules of procedure, committee jurisdiction, and procedural voting (including filibusters and cloture) play an independent role in determining the legislative outcome apart from voting power. Indeed, a parliamentarian with small Banzhaf power may wield enormous sway by controlling the agenda.

Lastly, these approaches provide no answers to the issue of institutional design. For example, in trade-robust voting systems, how does one choose from infinitely many choices of weights which

are all valid? The characterisation theorem shows that there are possible choices of weights but says nothing about the selection among them.

Methodology

This dissertation employs a formal-theoretic research design grounded in cooperative game theory, translating the constitutional and charter provisions of three institutions the United Nations Security Council (UNSC), the International Monetary Fund (IMF), and India's Goods and Services Tax (GST) Council into weighted voting games, and computing the Banzhaf Power Index and the Shapley-Shubik Index for each. The methodological choice reflects the nature of the research question: voting power, properly understood, is a property of institutional design rather than of political will, and its rigorous measurement demands formal mathematical tools.

Weighted Voting Games

Each institution is formally represented as a weighted voting game, denoted $[q : w_1, w_2, \dots, w_n]$, where q is the quota the minimum combined weight required for a coalition to be winning — and each w_i is the numerical weight assigned to player i . A coalition S is winning if and only if the sum of its members' weights meets or exceeds the quota. The UNSC is modelled as $[39 : 7, 7, 7, 7, 7, 1, 1, 1, 1, 1, 1, 1, 1]$, where permanent members hold weight 7 and non-permanent members weight 1, with the quota derived from the requirement that all five permanent members plus at least four non-permanent members must concur. The IMF is treated as a four-bloc game $[50 : 16.49, 27.52, 14.15, 41.85]$ under a simple majority threshold. The GST Council is represented as $[70 : 31, 2, 2, \dots, 2]$, with the Centre holding weight 31 and each of 31 States holding weight 2, under a quota of 70 derived from the constitutional three-quarters supermajority requirement.

The Banzhaf Power Index

The Banzhaf Power Index (Banzhaf, 1965) operationalises power as the frequency with which a player is critical that is, their removal converts a winning coalition into a losing one. Formally,

player i is critical in coalition S if $\sum_{j \in S} w_j \geq q$ but $\sum_{j \in S \setminus \{i\}} w_j < q$. The Total Banzhaf Power (TBP) of player i , denoted η_i , counts all such instances across the full coalition space. The normalised index is $BI(i) = \eta_i / \sum_j \eta_j$, distributing power as shares summing to unity. In large games such as the GST Council, exhaustive enumeration across 2^{32} coalitions is computationally prohibitive; closed-form binomial expressions are therefore employed. The Centre's TBP equals $\sum C(31, k)$ for $k = 20$ to 31 , and each State's TBP equals $C(30, 19)$, since a State is critical only in coalitions of exactly 20 States – the minimum threshold for the Centre to prevail.

The Shapley-Shubik Index

The Shapley-Shubik Index (Shapley and Shubik, 1954) derives from the Shapley value in cooperative game theory. It models coalition formation as a sequential process: all $n!$ orderings of players are considered with equal probability, and for each ordering the pivotal player is identified as the one whose entry causes the cumulative weight to first meet or exceed the quota. The index for player i is $SSI(i) = (\text{number of orderings in which } i \text{ is pivotal}) / n!$. Unlike the Banzhaf Index, the SSI is sensitive to the position in the formation sequence at which a player becomes decisive, capturing scenarios in which other players have already mobilised before the pivotal player enters.

Distinguishing the Two Indices

The two indices embody distinct normative assumptions. The Banzhaf Index treats all coalitions as equally probable and counts raw critical detections, making it apt for settings where coalitions form simultaneously. The Shapley-Shubik Index weights orderings equally, better capturing sequential bargaining dynamics. Their divergence – most pronounced in the GST Council, where the Centre's Banzhaf share of 89.74% far exceeds its Shapley-Shubik share of 37.5% – is itself analytically significant, revealing how structural dominance varies with assumptions about coalition formation.

Limitations

Both indices assume voter independence and assign equal probability to all coalitions or orderings, abstracting away from party discipline, ideological alignment, and strategic behaviour. The four-bloc IMF simplification sacrifices granularity for tractability. Neither index captures agenda-setting authority or informal diplomatic influence. These limitations are inherent to the power index methodology and are acknowledged throughout the literature; within their defined domain the formal distribution of decisive voting power under given institutional rules the Banzhaf and Shapley-Shubik indices remain the most rigorous instruments available.

Voting Power Analysis of the United Nations Security Council

INTRODUCTION AND PRIMARY RESPONSIBILITY OF THE UNSC

The United Nations Security Council (UNSC) is one of the key bodies of the United Nations, which deals with matters related to international peace and security. The UNSC was founded on 26 June 1945, with the signing of the UN Charter and is an organisational structure that represents the international understanding reached following World War II about the necessity of such a body with authority to impose decisions that prevent future wars and secure international peace and security (United Nations, 1945). This task is incumbent upon all UN members. Members of the UN Charter agree to adhere to the decisions taken by the Council according to the Charter of the United Nations. The distinguishing feature of the UNSC is its binding nature, unlike other UN organs, which give recommendations only.

MEMBERSHIP STRUCTURE AND COMPOSITION

The composition of the Security Council is arranged in such a manner that its members include great powers as well as the rest of the international community. The Security Council is composed of fifteen members of the UN that can be classified into two categories: five permanent members and ten non-permanent members (United Nations, 1945).

Permanent Members (P-5)

There are five permanent members of the UN Security Council: China, France, the Russian Federation, the United Kingdom of Great Britain and Northern Ireland, and the United States of America. These five nations have been considered permanent members due to their status as great Allied Powers after the end of the Second World War. There have been only two changes within the Security Council, namely, in 1971 when the seat of the Republic of China

was substituted by that of the People's Republic of China, and in 1992 after the collapse of the USSR when the seat of the latter was taken by Russia.

Non-Permanent Members

Furthermore, the General Assembly selects ten non-permanent members of the Security Council to serve a term of two years. While selecting these non-permanent members, the General Assembly takes into consideration the role played by the member in promoting and maintaining international peace and security, along with other undertakings of the member in relation to the United Nations, keeping into view the requirement of geographical representation.

THE VOTING SYSTEM: STRUCTURE AND MECHANICS

The Fundamental Voting Principle

Each member of the Security Council shall have one vote. While this principle ensures democratic equality among all fifteen members of the body, there is still a great level of intricacy associated with the process of voting. This intricacy arises from differences in procedural and substantive voting rules.

a) Procedural Issues

When it comes to procedural issues, the voting process in the Security Council must involve an affirmative vote from at least nine members of the council. The most important thing to note here is that the permanent members of the UN Security Council have no right to veto decisions when it comes to procedural voting. When it comes to procedural issues in the voting process, an adverse vote from permanent members will not suffice to block a procedural issue because at least nine positive votes are needed. Examples of such issues include adoption of agenda, inclusion of non-members during the discussion, among others.

b) Substantive Matters:

The issue of voting on substantive issues in the UN Security Council was discussed in article 27(3) of chapter V of the UN Charter, stating that decisions by the Security Council on any other question should involve an affirmative vote by nine members, including the agreement of the permanent members. For decisions in Chapter VI and paragraph three of Article 52, however, the voting parties should abstain from voting.

Veto Power Mechanism

The veto power is the main component in the UNSC's voting process (United Nations Security Council, 2025). Even though "veto" is not mentioned in the UN Charter, the mechanism of the veto originates from Article 27 of the UN Charter, which outlines the process of voting in the UNSC.

How Veto Works?

The veto power is implemented with a simple dissenting vote on any issue in the UNSC. If the permanent member casts a negative vote against the resolution being passed, then all other affirmative votes become worthless, regardless of the number of votes that favour the resolution. Thus, if one out of the 15 votes is negative, 14 affirmative votes won't count. This system is designed to safeguard the major powers from making any move against their best interest.

It should be noted that the power of veto applies to substantive issues and does not cover any procedural questions. Furthermore, an abstained or non-attended vote of a permanent member cannot serve as a veto for a decision. This distinction has enabled permanent members to allow resolutions to pass while signalling their non-support through abstention, providing a nuanced middle ground between full support and complete obstruction.

Introduction to Weighted Voting Systems

We are going to convert the UNSC into a weighted voting system; a formal framework in which different participants (players) are assigned distinct numerical weights that reflect their relative influence in collective decision-making. This framework is formally denoted as:

$$[q : w_1, w_2, \dots, w_n]$$

where each component of the notation carries a specific technical meaning:

Symbol	Definition
q	The quota; the minimum total weight required for a resolution to pass
w_i	The weight assigned to the i th player
n	The total number of players in the voting body
W	The sum of all individual weights (total weight)

In the context of the UNSC, the voting body is composed of 15 member states: 5 Permanent Members (P5) and 10 Non-Permanent Members (NP). A critical structural feature of this system is that all non-permanent members are assigned equal voting weight of 1, while permanent members carry a significantly higher weight, which is derived analytically below.

Structural Composition and Voting Rules of the UNSC

The UNSC requires a minimum of 9 affirmative votes to pass a resolution. Crucially, all 5 permanent members must vote in favour this is the well known veto power. A single permanent member voting against the resolution is sufficient to block it, regardless of the support from non-permanent members.

From this rule, the following structural implication follows directly:

- 5 permanent members all vote YES $\rightarrow$ contributes $5 \times P5$ weight to the total
- At minimum 4 non permanent members must also vote YES to reach the quota of 9
- Therefore, a minimum of 4 non-permanent member votes are required to pass any resolution

Derivation of Voting Weights and Quota

Setting Up the System of Inequalities

To determine the appropriate weight for a permanent member (x) and establish a consistent quota (q), two simultaneous conditions must hold:

Condition (i) A winning coalition (all P5 + 4 NP members) must meet the quota:

$$5x + 4 \geq q$$

Condition (ii) A losing coalition (all P5 + only 3 NP members, missing veto threshold) must fall short:

$$4x + 10 < q \rightarrow \text{will not pass}$$

Solving the System

Subtracting Condition (ii) from Condition (i) to eliminate q :

$$(5x + 4) \geq q > (4x + 10)$$

$$5x + 4 > 4x + 10$$

$$x > 6$$

From the second inequality, rearranging to find the upper bound:

$$4x + 10 < q \Rightarrow x < 6 + \epsilon$$

Setting $x = 7$ satisfies both conditions ($x > 6$ is satisfied since $7 > 6$).

Verification with $x = 7$:

- $5(7) + 4 = 39 \geq q$

- $4(7) + 10 = 38 < 39$

Therefore, setting $q = 39$ is consistent with both conditions.

Summary of Weights and Quota

Parameter	Value
Weight of each Permanent Member (P5)	7
Weight of each Non-Permanent Member (NP)	1
Quota (q)	39
Total Weight (W) = $5(7) + 10(1)$	45

Banzhaf Power Index

Definition

The Banzhaf Index measures the proportion of winning coalitions in which a player is critical i.e., their removal would cause the coalition to lose. It is defined as:

$$BI(P_i) = \text{Total Banzhaf Power of } P_i / \text{Sum of all Total Banzhaf Powers}$$

Identifying Winning Coalitions

A winning coalition in the UNSC must contain all 5 permanent members plus at least 4 non-permanent members (to satisfy the quota of 39). Formally:

$$\text{Winning Coalition} = 5P + k \times NP, \text{ where } k \geq 4$$

The number of ways to form such coalitions is determined by choosing k non-permanent members from 10:

${}^{10}C_4$	${}^{10}C_5$	${}^{10}C_6$	${}^{10}C_7$	${}^{10}C_8$	${}^{10}C_9 + {}^{10}C_{10}$
210	252	210	120	45	$10 + 1 = 11$

$$\text{Total Winning Coalitions} = 210 + 252 + 210 + 120 + 45 + 10 + 1 = 848$$

Banzhaf Power of Permanent Members

Each permanent member is critical in every single one of the 848 winning coalitions, since removing any P5 member from any winning coalition causes it to fall below the required quota.

$$\text{TBP(P5)} = 848$$

Banzhaf Power of Non-Permanent Members

A non-permanent member is critical only in coalitions where the coalition has exactly 4 NP members. In such a coalition, removing that NP member leaves only 3 NP members, which means the total weight drops to $35 + 3 = 38 < 39$, causing the coalition to lose.

The number of such critical detections for one NP member: choose the other 3 NP members from the remaining 9:

$$\text{Critical detections per NP} = {}^9C_3 = 84$$

$$\text{TBP(NP)} = 84$$

Total Banzhaf Power and Index Calculation

$$\begin{aligned}\text{Total TBP} &= \text{TBP(P5)} \times 5 + \text{TBP(NP)} \times 10 \\ &= 848 \times 5 + 84 \times 10 = 4240 + 840 = 5080\end{aligned}$$

The Banzhaf Index for each player type is then:

$$\text{BI(P5)} = 848 / 5080 = 212/1270 = 106/635 \approx 0.1669 \text{ (or 16.69\%)}$$

$$\text{BI(NP)} = 84 / 5080 = 21/1270 \approx 0.01654 \text{ (or 1.654\%)}$$

Verification

The sum of all Banzhaf indices must equal 1:

$$\begin{aligned}&5 \times (212/1270) + 10 \times (21/1270) \\ &= 1060/1270 + 210/1270 = 1270/1270 = 1\end{aligned}$$

Summary and Comparative Analysis

Measure	Permanent Member (P5)	Non-Permanent Member (NP)
Voting Weight	7	1
Quota (q)	39	39
Banzhaf Index	16.69%	1.654%
Critical Coalitions	848	84

It is evident that there exists a substantial discrepancy between the voting power of permanent and non-permanent members in the UNSC. Although the former has a voting weight of 7 while the latter has 1, their effective voting power using both SSI and BPI is significantly higher. Every permanent member holds about 19.62% of the pivotal power in the SSI method and 16.69% in the BPI method, whereas each of the non-permanent members holds less than 2% of the pivotal power in either case.

This is due to the veto power granted to the permanent members, meaning that they must be included in every winning coalition because they have a say in each of the results. However, the non-permanent members only hold pivotal positions in a limited number of coalitions, where their marginal votes determine the result at the edge of the quota limit.

Policy Recommendations: Structural Reform of the UNSC Voting System

The preceding analysis demonstrated that the formal architecture of the United Nations Security Council produces a profound asymmetry in actual voting power. Each permanent member (P5) commands a Banzhaf Index of **16.69%**, against each non-permanent member's (NP) **1.65%**, a tenfold disparity arising not merely from geopolitical status but from the mathematical structure of the veto mechanism itself. This section evaluates a structural reform through the same Banzhaf framework, deriving new power index computations to quantify the redistributive consequences of the proposal: **Reform — Double Veto Threshold:** Requiring at least two P5 members to oppose a resolution before it is blocked.

The Double Veto Threshold

Structural Motivation

Under the current system, every P5 member is critical in **every single one** of the 848 winning coalitions. This follows from the unanimity requirement on P5 participation: no resolution can pass unless all five permanent members concur, making the marginal contribution of each P5 vote absolute. The single-member veto thus produces a game-theoretic lock that enables unilateral obstruction structurally, each P5 member alone constitutes a minimal blocking coalition.

The double veto reform dissolves this individual blocking power. Under the proposed rule, a resolution is defeated only if **at least two** P5 members vote against it. A single negative P5 vote is insufficient to block. Procedurally, a P5 member wishing to veto would be required to convince at least one other permanent member to join them, reintroducing a deliberative check within the P5 itself.

Formal Definition of the Voting Game

The double veto system is defined by two simultaneous conditions. A coalition wins if and only if:

- **Condition (a):** The total number of affirmative votes is at least 9 (the existing nine-vote threshold is retained).
- **Condition (b):** At most one P5 member votes against equivalently, at least four P5 members vote in favour ($p \geq 4$).

This game is not representable as a single weighted voting system using the original weight assignment of $[39 : 7, 7, 7, 7, 7, 1, 1, \dots, 1]$. Under those weights, a coalition of four P5 members and all ten NP members yields a total weight of $4(7) + 10(1) = 38$, which falls short of the quota of 39. A new weight assignment consistent with both conditions must therefore be derived, or the game must be analysed combinatorially.

The combinatorial approach is adopted here, as it most faithfully captures the two-condition winning rule without imposing a potentially distorting single weight representation.

Identifying the Winning Coalitions

A winning coalition under the double veto rule consists of p P5 members and k NP members satisfying $p + k \geq 9$ and $p \geq 4$. This restricts p to the values $\{4, 5\}$.

For $p = 4$: the condition $k \geq 5$ must hold (since $4 + k \geq 9$ requires $k \geq 5$), with k ranging from 5 to 10. The number of coalitions of each type is $C(5, 4) \times C(10, k)$, reflecting the choice of 4 P5 members from 5 and k NP members from 10.

For $p = 5$: the condition $k \geq 4$ must hold, with k ranging from 4 to 10. Since all five P5 members participate, the count per NP configuration is $C(5, 5) \times C(10, k) = C(10, k)$.

Table 1 enumerates all winning coalitions under the double veto rule.

P5 Members (p)	NP Members (k)	Ways to Choose P5	Ways to Choose NP	Coalition Count
4	5	$C(5, 4) = 5$	$C(10, 5) = 252$	1,260
4	6	$C(5, 4) = 5$	$C(10, 6) = 210$	1,050
4	7	$C(5, 4) = 5$	$C(10, 7) = 120$	600
4	8	$C(5, 4) = 5$	$C(10, 8) = 45$	225
4	9	$C(5, 4) = 5$	$C(10, 9) = 10$	50
4	10	$C(5, 4) = 5$	$C(10, 10) = 1$	5
5	4	$C(5, 5) = 1$	$C(10, 4) = 210$	210
5	5	$C(5, 5) = 1$	$C(10, 5) = 252$	252
5	6	$C(5, 5) = 1$	$C(10, 6) = 210$	210
5	7	$C(5, 5) = 1$	$C(10, 7) = 120$	120
5	8	$C(5, 5) = 1$	$C(10, 8) = 45$	45
5	9	$C(5, 5) = 1$	$C(10, 9) = 10$	10
5	10	$C(5, 5) = 1$	$C(10, 10) = 1$	1
TOTAL				4,038

Banzhaf Power of a Permanent Member

A P5 member is critical in a winning coalition if and only if their removal converts the coalition from winning to losing. Two distinct cases arise:

Case	Coalition Type	Why P5 Member is Critical	k Range
Case A	$p = 4$	$p - 1 = 3 < 4$ (double-veto violated)	$k = 5$ to 10
Case B	$p = 5, k = 4$	$(p-1) + k = 4 + 4 = 8 < 9$ (vote threshold)	$k = 4$ only

Case A: Coalitions with exactly 4 P5 members ($p = 4$)

When $p = 4$, removing any one P5 member reduces the count to $p = 3$, violating Condition (b) (since $3 < 4$). The resulting coalition is therefore losing regardless of the NP count. Accordingly, **every P5 member is critical in all coalitions with $p = 4$** . The number of such coalitions containing a specific P5 member is obtained by choosing the remaining 3 P5 members from the other 4, multiplied by $C(10, k)$ for each valid k :

k (NP members)	Expression	Calculation	Count per P5
5	$C(4,3) \times C(10,5)$	4×252	1,008
6	$C(4,3) \times C(10,6)$	4×210	840
7	$C(4,3) \times C(10,7)$	4×120	480
8	$C(4,3) \times C(10,8)$	4×45	180
9	$C(4,3) \times C(10,9)$	4×10	40
10	$C(4,3) \times C(10,10)$	4×1	4
Case A Total	2,552		

Case A total per P5 member = 2,552

Case B: Coalitions with exactly $p = 5, k = 4$

When all five P5 members are present and exactly four NP members support the resolution ($k = 4$), the total is $5 + 4 = 9$, satisfying the threshold exactly. Removing one P5 member reduces the count to

$4 + 4 = 8 < 9$, violating Condition (a). Hence the P5 member is critical. The number of such coalitions containing a specific P5 member is $C(4, 4) \times C(10, 4) = 1 \times 210 = 210$.

Case B total per P5 member = 210

For all other winning coalitions ($p = 5, k > 4$), removing one P5 member yields $p = 4, k > 4$, which satisfies both conditions ($4 + k > 9$ and $4 \geq 4$). The coalition remains winning; the P5 member is **not critical**.

TBP(P5) = 2,552 + 210 = 2,762 per permanent member

Banzhaf Power of a Non-Permanent Member

A NP member is critical only when their removal causes the coalition to fall below the nine-vote threshold, i.e., when $p + k = 9$ exactly. The P5 composition does not affect criticality for NP members, since NP removal does not change p . The two relevant configurations, and the number of coalitions in which each NP member is critical, are:

p	k (incl. this NP)	Expression	Calculation	Count
4	5	$C(5,4) \times C(9,4)$	5×126	630
5	4	$C(5,5) \times C(9,3)$	1×84	84
TBP per NP member				714

TBP(NP) = 630 + 84 = 714 per non-permanent member

Banzhaf Index Computation

Total TBP (P5 bloc): $5 \times 2,762 = 13,810$

Total TBP (NP bloc): $10 \times 714 = 7,140$

Grand Total TBP: 20,950

$$BI(P5) = 2,762 / 20,950 = 1,381/10,475 \approx 0.1318 \text{ (13.18\%)}$$

$$BI(NP) = 714 / 20,950 = 357/10,475 \approx 0.0341 \text{ (3.41\%)}$$

Verification: $5 \times 0.1318 + 10 \times 0.0341 = 0.659 + 0.341 = 1.000$

Player	TBP	Banzhaf Index (fraction)	Banzhaf Index (%)
P5 Member (each)	2,762	2,762 / 20,950	13.18%
NP Member (each)	714	714 / 20,950	3.41%
P5 Bloc (all 5)	13,810	13,810 / 20,950	65.92%
NP Bloc (all 10)	7,140	7,140 / 20,950	34.08%

Redistribution of Power and Institutional Implications

Double veto generates a considerable redistribution of the Banzhaf Index. The value of the individual P5 member's Banzhaf Index declines from 16.69% to 13.18%, which constitutes a reduction of 3.51 percentage points for each of the five countries. What is even more striking is the change in the value of the individual NP Banzhaf Index, which grows more than twice from 1.65% to 3.41%.

In other words, what happens here is an absolutely clear-cut redistribution process: with the double veto applied, there will be numerous cases where four members out of five (or more specifically, four out of four, since all five members need at least four to generate their respective coalitions) will form a coalition in which each of the four members is necessary and cannot leave the coalition and veto the decision made without being able to recruit another P5 member.

As expected, the power of the NP bloc increases from 16.54% to 34.08%. It demonstrates the formation of a broader range of winning coalition possibilities, where the representatives of the NP group perform critical functions. The total number of winning coalitions goes up from 848 to 4,038,

thus expanding the potential area of coalition by almost five times. Consequently, the number of opportunities to perform as a critical voter for either P5 or NP member grows significantly.

Institutionally, the reform sustains the idea of the great powers' consensus but abolishes the ability to obstruct the decision-making process individually. In other words, the Council would not be able to stall its decision-making process because of a single P5 member anymore; the state would need to convince another P5 state to block the proposal, too. Therefore, the process of using veto power is more consistent with the initial intention of the UN Charter concerning the joint actions of great powers for maintaining global peace and order.

Voting Power Analysis of the International Monetary Fund

Introduction to the International Monetary Fund

The International Monetary Fund (IMF) is arguably one of the top-ranking multilateral financial organisations globally, with its primary objectives being to ensure international monetary cooperation, financial stability, support international trade, promote sustainable economic growth, and alleviate poverty among its member countries. The IMF came into being during the famous Bretton Woods Conference held in New Hampshire, USA, in 1944 and started functioning from 1947 after its official creation in 1945. The idea behind the creation of this institution was inspired by the dire effects of World War II, especially those caused by the Great Depression, competitive currency devaluations, and failure of international trade that occurred due to the lack of international economic cooperation.

Firstly, the main responsibility of the IMF involves the supervision of the international monetary system, as well as overseeing economic and financial policies of member countries. In cases when countries have balance-of-payments problems and cannot fulfil their international financial commitments, the IMF can provide financial aid through loans and credits. Beyond crisis lending, the IMF plays a central role in global economic surveillance, analysing national, regional, and global economic trends and providing policy recommendations through flagship publications such as the World Economic Outlook and the Global Financial Stability Report.

As of the time of writing, the IMF has 191 member countries, and hence, is a quasi-universal organisation in terms of membership in the world community. Every member receives a quota that indicates the country's relative importance within the context of global economic relations and serves as a basis for determining the amount of its contribution to the Fund's resources and its voting

rights in the Fund. Hence, the IMF functions according to the scheme of weighted voting when the weight of voting rights depends on the country's importance from the perspective of its economic contribution to the global economy. The U.S. with the biggest economy receives approximately 16.49% of voting shares and effectively holds a veto in some cases, requiring a supermajority of 85%.

The IMF plays a key role not only in the management of crises but in international economic policy making through dialogue and exchange of ideas, provision of information and analysis, and assistance in capacity building for poorer members of the organisation. Therefore, the design of IMF in the sense of voting system becomes relevant for further analysis.

IMF Structure and Voting Framework

Membership and Vote Allocation

The IMF has 191 member countries. Unlike the UNSC, where veto power is explicitly granted to five permanent members, the IMF allocates voting shares as percentages calculated by the Fund itself based on each member's quota, which reflects its GDP, openness, variability of current accounts, and international reserves. Resolutions may require different majority thresholds depending on their nature.

For the purposes of this analysis, the 191 members are aggregated into four representative blocs to simplify the calculation of power indices while preserving the essential structural dynamics of the institution:

Bloc	Members	Voting Share
US	United States	16.49%
EU	31 European countries (see Appendix)	27.52%
BRICS	China, India, Russia, Brazil, South Africa	14.15%

RoW	Rest of the World (remaining 154 members)	41.85%
Total		100%

The Rest of the World (RoW) share is derived residually:

$$\text{RoW} = 100 - [16.49 + 27.52 + 14.15] = 41.85\%$$

EU Bloc Composition and Voting Shares

The EU bloc comprises 31 European countries whose IMF vote shares aggregate to 27.52%. The individual country shares are as follows:

Country	Share (%)	Country	Share (%)	Country	Share (%)
Germany	5.31	Ireland	0.71	Croatia	0.17
France	4.03	Denmark	0.71	Slovenia	0.15
Italy	3.02	Greece	0.51	Lithuania	0.12
Spain	1.92	Finland	0.51	Latvia	0.09
Netherlands	1.76	Czechia	0.46	Cyprus	0.09
Belgium	1.30	Portugal	0.44	Estonia	0.08
Switzerland	1.17	Hungary	0.41	Malta	0.06
Sweden	0.91	Romania	0.34	Liechtenstein	0.05
Poland	0.84	Luxembourg	0.29		–
Austria	0.81	Slovakia	0.23		–
Norway	0.77	Bulgaria	0.21	EU Total	27.52%

BRICS Bloc Composition

The BRICS bloc comprises 5 countries with a combined voting share of 14.15%:

Country	Voting Share (%)
China	6.08

Russia	2.84
India	2.63
Brazil	2.22
South Africa	0.63
BRICS Total	14.15%

Coalition Analysis and Majority Threshold

Winning Threshold

To pass a resolution at the IMF, a coalition must command more than 50% of the total votes (for standard resolutions). Some decisions, such as quota increases and SDR allocations, require a supermajority of 85%, which effectively grants the United States a veto. For this analysis, the 50% threshold is used unless otherwise stated.

Given the four-bloc model, all 16 possible coalition subsets (including the empty set) are evaluated. The table below lists all coalitions, their combined vote share, and whether each constitutes a winning coalition (W) or a losing coalition (L):

#	Coalition	Combined Share (%)	>50%?	Result
1	US	16.49	No	L
2	EU	27.52	No	L
3	BRICS	14.15	No	L
4	RoW	41.85	No	L
5	US + EU	44.01	No	L
6	US + BRICS	30.64	No	L
7	US + RoW	58.34	Yes	W
8	EU + BRICS	41.67	No	L
9	EU + RoW	69.37	Yes	W
10	BRICS + RoW	56.00	Yes	W

11	US + EU + BRICS	58.16	Yes	W
12	US + EU + RoW	85.86	Yes	W
13	US + BRICS + RoW	72.49	Yes	W
14	EU + BRICS + RoW	83.52	Yes	W
15	US + EU + BRICS + RoW	100.00	Yes	W
16	Empty Coalition	0.00	No	L

Out of 16 possible subsets, 8 are winning coalitions. The four single-bloc coalitions, the two-bloc coalitions without RoW or without US+EU, all fall short of the 50% threshold.

Banzhaf Power Index

Methodology

The Banzhaf Power Index (BI) measures a bloc's criticality the extent to which its presence in a coalition is decisive. A player is critical to a winning coalition if, upon their removal, the coalition becomes losing. The index is defined as:

$$BI(P) = TBP(P) / \text{Sum of all TBP}$$

where TBP (Total Banzhaf Power) counts the number of winning coalitions in which player P is critical.

For each winning coalition, we assign +1 to each bloc that is present and −1 to each bloc that is absent. A player's critical score in that coalition is assigned only if flipping their status (from +1 to −1) would change the coalition from winning to losing.

Critical Detection Table

The table below shows the sign (+1 = in coalition, −1 = not in coalition) for each bloc across all winning coalitions, and identifies the critical player in each:

Winning Coalition	US	EU	BRICS	RoW
US + RoW	+1	-1	-1	+1
EU + RoW	-1	+1	-1	+1
BRICS + RoW	-1	-1	+1	+1
US + EU + BRICS	+1	+1	+1	-1
US + EU + RoW	+1	+1	-1	+1
US + BRICS + RoW	+1	-1	+1	+1
EU + BRICS + RoW	-1	+1	+1	+1
US + EU + BRICS + RoW	+1	+1	+1	+1

Analysing each winning coalition for critical blocs:

- US + RoW: Both US and RoW are critical (removing either drops below 50%)
- EU + RoW: Both EU and RoW are critical
- BRICS + RoW: Both BRICS and RoW are critical
- US + EU + BRICS: All three are critical (removing any one causes the coalition to lose)
- US + EU + RoW: US and EU are not individually critical (EU+RoW alone = 69.37% > 50%); only examining marginal cases here US is NOT critical (EU+RoW wins), EU is NOT critical (US+RoW wins), RoW is NOT critical (US+EU = 44% < 50% so RoW is critical)
- US + BRICS + RoW: US is not critical (BRICS+RoW = 56% > 50%); BRICS is not critical (US+RoW = 58.34% > 50%); RoW IS critical (US+BRICS = 30.64% < 50%)
- EU + BRICS + RoW: EU is not critical (BRICS+RoW = 56% > 50%); BRICS is not critical (EU+RoW = 69.37% > 50%); RoW is not critical (EU+BRICS = 41.67% < 50% so RoW is critical)
- US + EU + BRICS + RoW: No bloc is critical (any three bloc coalition still wins)

Tallying the Total Banzhaf Power (TBP) for each bloc:

Bloc	TBP	Fraction	BI (%)
US	2	2/12	16.67%

EU	2	2/12	16.67%
BRICS	2	2/12	16.67%
RoW	6	6/12	50.00%
Total	12	12/12	100%

The Banzhaf Index results reveal a striking paradox: despite holding only 41.85% of the vote share, the Rest of the World bloc commands 50% of the Banzhaf power three times the power of the US, EU, or BRICS individually. This arises because RoW is critical to the greatest number of winning coalitions, as the other three blocs all require RoW's participation to reach the 50% threshold in most configurations.

It is important to note that these results are a product of the four-bloc simplification. In reality, the 191 IMF members each hold distinct vote shares, and the full computation of power indices would require enumeration across all possible coalitions among 191 players, a computationally intensive exercise. The bloc model is a pedagogically useful approximation that nonetheless captures the essential institutional dynamics of the Fund's governance structure.

Shapley-Shubik Power Index (SSI)

Definition and Setup

The Shapley-Shubik Index (SSI) measures the probability that a player is the pivotal member in a randomly ordered sequential coalition the player whose entry causes the cumulative vote share to cross the winning threshold of 50%. With $n = 4$ blocs, the total number of orderings is:

$$4! = 24 \text{ orderings}$$

The voting shares used throughout the SSI analysis are:

- US: 16.49%
- EU: 27.52%
- BRICS: 14.15%

- RoW: 41.85%

5.2 All 24 Orderings and Pivotal Players

Each of the 24 permutations of the four blocs is evaluated by tracking the cumulative vote share as blocs enter sequentially. The bloc whose entry causes the running total to first exceed 50% is identified as the pivotal player.

#	Ordering	Cumulative Shares	Pivot
1	US → EU → BRICS → RoW	16.49 → 44.01 → 58.16 → 100	BRICS
2	US → EU → RoW → BRICS	16.49 → 44.01 → 85.86 → 100	RoW
3	US → BRICS → EU → RoW	16.49 → 30.64 → 58.16 → 100	EU
4	US → BRICS → RoW → EU	16.49 → 30.64 → 72.49 → 100	RoW
5	US → RoW → EU → BRICS	16.49 → 58.34 → 85.84 → 100	RoW
6	US → RoW → BRICS → EU	16.49 → 58.34 → ...	RoW
7	EU → US → BRICS → RoW	27.52 → 44.01 → 58.16 → 100	BRICS
8	EU → US → RoW → BRICS	27.52 → 44.01 → 85.86 → 100	RoW
9	EU → BRICS → US → RoW	27.52 → 41.67 → 58.16 → 100	US
10	EU → BRICS → RoW → US	27.52 → 41.67 → 83.52 → 100	RoW
11	EU → RoW → US → BRICS	27.52 → 69.37 → ...	RoW
12	EU → RoW → BRICS → US	27.52 → 69.37 → ...	RoW
13	BRICS → US → EU → RoW	14.15 → 30.64 → 58.16 → 100	EU
14	BRICS → US → RoW → EU	14.15 → 30.64 → 72.49 → 100	RoW
15	BRICS → EU → US → RoW	14.15 → 41.67 → 58.16 → 100	US
16	BRICS → EU → RoW → US	14.15 → 41.67 → 83.52 → 100	RoW
17	BRICS → RoW → US → EU	14.15 → 56.00 → ...	RoW
18	BRICS → RoW → EU → US	14.15 → 56.00 → ...	RoW
19	RoW → US → EU → BRICS	41.85 → 58.34 → ...	US
20	RoW → US → BRICS → EU	41.85 → 58.34 → ...	US

21	RoW → EU → US → BRICS	41.85 → 69.37 → ...	EU
22	RoW → EU → BRICS → US	41.85 → 69.37 → ...	EU
23	RoW → BRICS → US → EU	41.85 → 56.00 → ...	BRICS
24	RoW → BRICS → EU → US	41.85 → 56.00 → ...	BRICS

5.3 Pivotal Count and SSI Results

Counting the number of orderings in which each bloc is pivotal:

Bloc	Pivotal Orderings	Ordering Numbers	SSI (%)
US	4	9, 15, 19, 20	4/24 = 16.67%
EU	4	3, 13, 21, 22	4/24 = 16.67%
BRICS	4	1, 7, 23, 24	4/24 = 16.67%
RoW	12	2,4,5,6,8,10,11,12,14,16,17,18	12/24 = 50.00%
Total	24	—	100%

The SSI results are consistent with the Banzhaf Index. RoW is the pivotal player in 12 out of 24 orderings (50%), while US, EU, and BRICS each account for exactly 4 orderings (16.67% each). This symmetry arises because US, EU, and BRICS are structurally symmetric in this four-bloc model none can reach 50% independently, and each requires RoW or both of the other two blocs to form a winning coalition.

Formally: $SSI(US) = SSI(EU) = SSI(BRICS) = 4/24 = 1/6 \approx 16.67\%$, and $SSI(RoW) = 12/24 = 1/2 = 50.00\%$.

6. Summary and Comparative Analysis

6.1 Vote Share vs. Power Indices

Bloc	Vote Share (%)	BI (%)	SSI (%)	Power / Vote Ratio
US	16.49	16.67	16.67	≈ 1.01
EU	27.52	16.67	16.67	≈ 0.61
BRICS	14.15	16.67	16.67	≈ 1.18
RoW	41.85	50.00	50.00	≈ 1.19

The analysis exposes a fundamental disjunction between formal vote share and effective voting power. The EU, despite holding the largest share of any single bloc at 27.52%, has the same power index as the US and BRICS each holding just 16.67% under both metrics. This is because the EU, like the other smaller blocs, cannot reach the winning threshold of 50% without forming coalitions, and in the four-bloc model, its structural position is equivalent to that of the US and BRICS.

The Rest of the World bloc emerges as the most powerful player under both indices (50%), despite not holding the largest nominal vote share. This result illustrates the power of swing voters coalitions that are structurally indispensable to the greatest number of winning configurations will always exhibit disproportionate power relative to their raw vote weight.

It is important to note that these results are a product of the four-bloc simplification. In reality, the 191 IMF members each hold distinct vote shares, and the full computation of power indices would require enumeration across all possible coalitions among 191 players a computationally intensive exercise. The bloc model is a pedagogically useful approximation that nonetheless captures the essential institutional dynamics of the Fund's governance structure.

Voting Power Analysis of the GST Council of India

INTRODUCTION TO THE GST COUNCIL

The Goods and Services Tax Council (GSTC) is an innovative institutional framework set up under Article 279A of the Constitution of India via the Constitution (One Hundred and First Amendment) Act, 2016. It has been described as one of the most revolutionary institutional frameworks adopted in India's fiscal federalism, where for the first time, a permanently constituted forum has been created that serves as a discussion and negotiation platform for making collective recommendations for India's indirect taxation framework involving not just the Union but also all State governments.

Before the introduction of GST, the indirect tax system prevailing in India was characterised by fragmentation in terms of multiple layers of indirect taxes levied either at the Central level (e.g., Central Excise Duty, Service Tax), or at the State level (VAT, Entry Tax, Octroi, among others). Such a tax system led to cascading effects, compliance complexities, and economic distortions in inter-state commerce. This made the need for an institutional arrangement for jointly deciding on a comprehensive indirect tax system very important.

Formally, the Council came into existence through a Presidential Notification on 12 September 2016, following the decision taken by the Union Cabinet on the same day. The Council is headed by the Union Finance Minister. It has a total of 33 voting members: two from the Central Government, namely, Union Finance Minister and Union Minister of State for Revenue/Finance, and thirty-one from State Governments and Union Territories having legislatures (28 States and 3 UTs, namely Delhi, Puducherry, and Jammu & Kashmir).

The responsibilities of the Council as laid down under Article 279A(4) include determination of goods and/or services to be subject to or exempted from GST; recommendation of GST rates and

their structure including floor rates with bands; formulation of model GST laws; laying down principles relating to place of supply, input tax credit, and threshold limits; and making special provision for any emergency like natural calamities, etc., and particular States. As of the year 2024, the Council had held 55 meetings and taken some important decisions including implementation of e-Way bill scheme (24th meeting); e-invoicing (35th meeting); rate rationalisation of electric vehicles (36th meeting); duty-free importation of raw materials and capital goods during the period of pandemic .

Despite the sweeping nature of its authority, the Supreme Court of India in *Union of India v. Mohit Minerals Pvt. Ltd.* (2022) clarified that the Council's recommendations are not legally binding on the Centre or the States, both Parliament and State Legislatures retain concurrent power under Article 246A to legislate on GST independently. This renders the Council a consensus-building body of immense political weight rather than a sovereign legislative authority, making the formal analysis of voting power within it all the more significant for understanding the dynamics of cooperative federalism in India.

VOTING STRUCTURE AND WEIGHTED VOTING SYSTEM

Constitutional Basis: Article 279A(9)

The voting rules governing the GST Council are prescribed under Article 279A(9) of the Constitution:

- Every decision of the Council shall be taken at a meeting by a majority of not less than three-fourths (75%) of the weighted votes of the members present and voting.
- The vote of the Central Government shall have a weightage of one-third (1/3) of the total votes cast in that meeting.
- The votes of all the State Governments taken together shall have a weightage of two-thirds (2/3) of the total votes cast in that meeting.

This creates a two-player weighted voting game in its simplest formulation: the Centre (C) and the States collectively (S). The quota is set at 75% of weighted votes. Translating to a normalised numerical framework for computation:

Numerical Weight Assignment

To make the voting game tractable for power index computation, we assign integer weights consistent with the constitutional proportions. The most natural normalisation sets:

Total normalised weight = 3 (for mathematical convenience)

Weight of Centre (C) = 1 [representing 1/3 of total votes]

Weight of States collectively (S) = 2 [representing 2/3 of total votes]

Quota (q) = $\frac{3}{4} \times 3 = 2.25$ -> in practice, q = 3 (since both must agree)

However, since the quota of 75% of 3 = 2.25, any winning coalition must have weighted votes ≥ 2.25 . In integer terms, this means a winning coalition must accumulate a total weight of at least 3, which is only achievable if both C and S are present. This is the key structural feature of the GST Council's voting game:

Coalition	C Weight	S Weight	Total	≥ 2.25 ?
Centre alone (C)	1	0	1	No (L)
States alone (S)	0	2	2	No (L)
Centre + States (C+S)	1	2	3	Yes (W)
Empty Coalition	0	0	0	No (L)

Out of 4 possible coalitions ($2^2 = 4$), only 1 is a winning coalition, the grand coalition of Centre and States together. Both the Centre and the States individually are losing coalitions, and critically, both are necessary for any resolution to pass.

This means the GST Council's voting game at its simplest level is a unanimity game of two players, a structure with profound implications for power analysis. Both players are equally critical in every winning coalition.

Reforming Game : Centre vs. Individual States

3.1 Distributing the States' Two-Thirds Weight Equally

The constitutional text specifies that States together hold 2/3 of the vote. However, it does not prescribe how this 2/3 is distributed among the individual States. As academic literature on this subject (notably Chakravarty et al. / MPRA Working Paper 86355) observes, a natural and widely-used approach for theoretical analysis is to assign equal votes to each State, since no constitutional provision or agreed formula distributes the 2/3 share differentially.

Under this assumption, with 31 State/UT members, each State receives an equal share of the 2/3 weight. For computational purposes, we convert to a clean integer representation:

Multiply all weights by 3 x 31 = 93

Centre's weight: $(1/3) \times 93 = 31$

Each State's weight: $(2/3) \times 93 / 31 = 2$ (each of the 31 states gets weight 2)

Total weight = $31 + (31 \times 2) = 31 + 62 = 93$

Quota = 75% of 93 = 69.75 -> $q = 70$ (rounding up, as votes must be integers)

The weighted voting game is therefore:

[70 : 31, 2, 2, 2, ..., 2] (Centre has weight 31; each of 31 States has weight 2)

This formulation is consistent with constitutional intent: the Centre alone (weight 31) falls far short of the quota (70); all 31 States together (weight 62) also fall short; only the Centre plus at least 20 or more States can reach the quota. Let us derive this threshold:

$$31 + 2k \geq 70$$

$$2k \geq 39$$

$$k \geq 19.5$$

$$k \geq 20$$

Therefore, the Centre requires the support of at least 20 out of 31 States to pass any resolution. Conversely, if 13 or more States oppose a measure while the Centre supports it, the resolution fails. This asymmetry is structurally significant: the Centre is a necessary but not sufficient player, while the States are sufficient as a blocking force even without any individual State holding veto power

State-Only Blocking Threshold

A coalition of States can block the Centre even without the Centre's participation only if their combined weight meets the threshold for a winning coalition on its own. Since the quota is 70 and the Centre holds 31, the States alone can reach a maximum of 62 which is less than 70. Therefore:

- States alone (all 31 combined) = $62 < 70$ -> The States collectively CANNOT pass a resolution without the Centre.
- Centre alone = $31 < 70$ -> The Centre CANNOT pass a resolution without the States.
- Both must cooperate for any resolution to pass, confirming the unanimity-game structure at the bloc level.

However, for blocking purposes: any coalition that prevents the complement from forming a winning coalition is a blocking coalition. Since the Centre is a necessary player in every winning coalition, the Centre alone constitutes a minimal blocking coalition, meaning the Centre holds a de facto veto over every resolution, consistent with the supermajority design.

Similarly, 13 or more States acting together can form a blocking coalition, since their combined weight (26) is enough to deny the Centre the 20-State support it needs:

If 13 States defect: Centre + 18 States = $31 + 13(2) = 67 < 70 \rightarrow$ Blocking coalition

Banzhaf Power Index

Methodology

The Banzhaf Power Index (BI) for a player is the ratio of the number of winning coalitions in which that player is critical (their removal causes the coalition to become losing) to the total critical detections across all players. The game is:

$[70 : 31, 2, 2, \dots, 2]$ — 1 Centre (C), 31 States each with weight 2

Banzhaf Power for the Centre

The Centre (weight 31) is critical in a winning coalition if removing the Centre causes it to fall below the quota. A winning coalition involving the Centre must satisfy:

$$31 + 2k \geq 70$$

$$k \geq 20$$

So winning coalitions containing the Centre have between 20 and 31 States. Therefore, the Centre is critical in EVERY winning coalition it participates in.

Number of winning coalitions containing the Centre (k ranges from 20 to 31):

$$\text{TBP(C)} = \text{Sum } [C(31, k)] \text{ for } k = 20 \text{ to } 31$$

k	C(31,k)	States Needed	Coalition Total Weight
20	84,672,315	20	31+40=71 $\geq$ 70
21	247,110,827	21	31+42=73 $\geq$ 70
22	600,805,296	22	31+44=75 $\geq$ 70
23	1,166,803,110	23	31+46=77 $\geq$ 70
24	1,855,967,520	24	31+48=79 $\geq$ 70
25	2,414,728,580	25	31+50=81 $\geq$ 70
26	2,629,575,880	26	31+52=83 $\geq$ 70
27	2,397,134,080	27	31+54=85 $\geq$ 70
28	1,855,967,520	28	31+56=87 $\geq$ 70
29	1,107,568,320	29	31+58=89 $\geq$ 70
30	465,421,215	30	31+60=91 $\geq$ 70
31	1	31	31+62=93 $\geq$ 70
Total TBP(C)	14,821,763,664		

Banzhaf Power for a Single State

A State (weight 2) is critical in a winning coalition if the coalition just barely reaches the quota, such that removing this State (reducing total by 2) would drop it below 70. For the Centre + k States, criticality of a single State requires:

$$31 + 2k \geq 70 \quad \text{AND} \quad 31 + 2(k-1) < 70$$

$$\text{First condition: } k \geq 20$$

$$\text{Second condition: } 31 + 2k - 2 < 70$$

$$2k < 41$$

$$k < 20.5$$

$$k \leq 20$$

Therefore: k = 20 exactly

A State is critical ONLY in coalitions where exactly 20 States (including itself) support the Centre.

To count these: choose the other 19 States from the remaining 30:

$$\text{TBP per State} = C(30, 19) = C(30, 11) = 54,627,300$$

This result is intuitive and powerful: a State is critical only in the razor-thin coalitions when 20 States back the Centre (exactly at the minimum threshold). In all coalitions with 21 or more States, no single State is individually critical, as removing one still leaves 20 States, which remains a winning coalition. This confirms that individual States have very limited pivotal power

Banzhaf Index Computation

$$\text{TBP(C)} = 14,821,763,664$$

$$\text{TBP per State} = 54,627,300$$

$$\text{TBP(all 31 States)} = 31 \times 54,627,300 = 1,693,446,300$$

$$\begin{aligned} \text{Total TBP} &= \text{TBP(C)} + \text{TBP(all States)} = 14,821,763,664 + 1,693,446,300 = \\ &16,515,209,964 \end{aligned}$$

Player	TBP	BI (fraction)	BI (%)
Centre (C)	14,821,763,664	$14,821,763,664 / 16,515,209,964$	89.74%
Each State (S _i)	54,627,300	$54,627,300 / 16,515,209,964$	0.331%
All 31 States	1,693,446,300	$1,693,446,300 / 16,515,209,964$	10.25%
Total	16,515,209,964	1	100%

Banzhaf power distribution indicates a great discrepancy between the Centre's power share and the individual States' shares. Thus, the Centre's power accounts for 89.74% of the total power in the

system, whereas each individual State holds just 0.331%. Such a significant disparity can be explained by the fact that the Centre carries significantly more weight (31) than each of the States (2) in the case of a high quota (70) and a total number of votes (93). In addition, the Centre is critical for 8.75 times more winning coalitions than all 31 States put together, because the Centre is indispensable for any winning coalition, but each State is critical in only marginal coalitions comprising 20 States.

Two Player Bloc Analysis

As noted in Section 2, the GST Council at its most fundamental level is a two player game: the Centre (C) and the States collectively (S). In this simplified game:

Coalition	Weight	$\geq$ Quota?	Result
Centre alone	$1/3 = 33.3\%$	No ($< 75\%$)	L
States alone	$2/3 = 66.7\%$	No ($< 75\%$)	L
Centre + States	$3/3 = 100\%$	Yes ($= 100\%$)	W
Empty	0	No	L

In this two-player interpretation, there is only 1 winning coalition (C+S). Both C and S are critical in that coalition. Therefore:

$$\text{TBP}(C) = \text{TBP}(S) = 1$$

$$\text{BI}(C) = \text{BI}(S) = 1/2 = 50\%$$

$$\text{SSI}(C) = \text{SSI}(S) = 1/2 = 50\% \quad [2 \text{ orderings: CS and SC; C pivotal in CS, S pivotal in SC}]$$

This two player analysis establishes an important baseline: when treating all States as a single unified bloc, both the Centre and States hold equal power (50% each) under both indices. This is the ideal of cooperative federalism encoded in Article 279A neither the Centre nor the States can govern

the GST framework unilaterally. The analysis of individual States then shows how this 50% State bloc power is disaggregated asymmetrically across 31 equal weight players.

Conclusion

In this study, voting power was analysed for three organisations with multilateral voting processes: the UN Security Council, the IMF, and the GST Council of India using formal cooperative game theory to calculate the real pivotal power distributions according to their current constitutions. A clear trend has emerged in all three organisations: formal voting power differs significantly from the power to influence decision making outcomes.

Analysis of voting power in the UNSC showed that the power imbalance created by the veto is extraordinary. Each permanent member has 16.69% of Banzhaf power, which is ten times as high as any non-permanent member, not because of the mandate prescribed in the Charter but because of necessity, as each of the five permanent members is a member of every winning coalition. Such design was supposed to maintain the great power consensus but evolved into an obstacle that explains the UNSC's paralysation when the great powers disagree. The double veto amendment may fix the problem.

IMF: Here, there is a paradox where "Rest of the World" has 41.85 percent votes but controls 50 percent Banzhaf power. As the swing voter bloc, the power depends on coalition structure rather than vote distribution. The 27.52 percent votes of the EU do not provide more power than the 14.15 percent votes of BRICS because both enjoy identical structural power status.

GST Council: It is the best example of deviation between formal voting weight and power distribution. While the Centre enjoys only 33.3 percent votes, it controls 89.74 percent Banzhaf power (37.5 percent according to Shapley-Shubik index). The Centre becomes more influential because of 75 percent super-majority vote requirement and its indispensability.

What These Findings Imply for Institutional Design:

These findings show that there are structural characteristics of voting procedures that allocate power independent of the intentions of those who designed them. It is not just the distribution of weights that determines influence; the threshold for decisions, the size of coalitions, and the use of veto procedures radically alter power allocations. Institutional designers should not take it for granted that proportional votes will equal proportional influence.

Democratic Consequences:

The study reveals profound democratic problems in weighted voting arrangements. There can be no legitimacy if there is a sharp gap between formal rules and substantive power. Power does not arise from design but through accidental structures. The power structure of the UNSC P5, the GST Centre, and the IMF swing voter are structural characteristics that cannot be attributed to constitutional design principles.

More General Implications:

One is that voting rules are mathematical constructs open to precise analysis; the workings of institutions need to be understood through rigorous study of their structures. Two is that swing voters have undue power, an insight that applies widely. Three is that supermajority requirements give extra weight to the supporters of the status quo and key players. Four is that discrepancies between the Banzhaf and Shapley-Shubik indices indicate that coalition assumptions are important; institutions should use more than one index.

Reform Recommendations:

Institutions need to be reformed structurally and not just behaviorally. Reforms like the double veto of the UNSC show how mathematics can guide such reforms. For the GST Council and the IMF, the above mathematical analysis shows what structural changes will be needed to make operations conform to its constitution.

All voting systems are not neutral mechanisms; they represent distributions of power which are either congruent or incongruent to the objectives of the institutions and democratic values. The current study has shown that an appreciation of the mechanisms of institutional operations, compared to merely knowing what they are supposed to do, involves the use of power indices. The glaring differences identified in three leading institutions indicate that international governance and federal governance at home take place under constitutions where implementation falls short of intention. Reducing this gap is a prerequisite of good institution building.

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